

# Advanced Terahertz Refractive Sensing And Fingerprint Recognition Through Metasurface-Excited Surface Waves

Zeyan Zhang, Zhuo Wang, Chao Zhang, Zhibo Yao, Shoujun Zhang, Ride Wang, Zhen Tian, Jiaguang Han, Chao Chang,\* Jing Lou,\* Xueqing Yan,\* and Chengwei Qiu\*

High-sensitive metasurface-based sensors are essential for effective substance detection and insightful bio-interaction studies, which compress light in subwavelength volumes to enhance light–matter interactions. However, current methods to improve sensing performance always focus on optimizing near-field response of individual meta-atom, and fingerprint recognition for bio-substances necessitates several pixelated metasurfaces to establish a quasi-continuous spectrum. Here, a novel sensing strategy is proposed to achieve Terahertz (THz) refractive sensing, and fingerprint recognition based on surface waves (SWs). Leveraging the long-range transmission, strong confinement, and interface sensitivity of SWs, a metasurface-supporting SWs excitation and propagation is experimentally verified to achieve sensing integrations. Through wide-band information collection of SWs, the proposed sensor not only facilitates refractive sensing up to  $215.5^\circ/\text{RIU}$ , but also enables the simultaneous resolution of multiple fingerprint information within a continuous spectrum. By covering  $5\ \mu\text{m}$  thickness of polyimide, quartz and silicon nitride layers, the maximum phase change of  $91.1^\circ$ ,  $101.8^\circ$ , and  $126.4^\circ$  is experimentally obtained within THz band, respectively. Thus, this strategy broadens the research scope of metasurface-excited SWs and introduces a novel paradigm for ultrasensitive sensing functions.

## 1. Introduction

The detection and sensing of biological analytes are intricately linked to human health and well-being. In domains such as disease surveillance,<sup>[1–3]</sup> cell identification,<sup>[4]</sup> and neural monitoring,<sup>[5]</sup> the accurate detection of bio-molecules is of paramount importance. High-sensitive sensors for specific bio-molecules provide a powerful approach for collecting biological information and assessing physiological processes, which lay the foundation in physiological research and human health detection.<sup>[2,5,6]</sup> Therefore, there is a growing demand to enhance the capability of distinguishing specific bio-molecules from large samples.<sup>[7,8]</sup> Such as in scenarios where vast samples are accessible, and immediate response time is not a necessity, conventional biological molecular detection techniques like polymerase chain reaction and immunoassay can still be effective, like virus sample detection.<sup>[9–11]</sup> As the requirement for improved bio-detection performance

Z. Zhang, C. Chang, X. Yan  
School of Physics  
Peking University  
Beijing 100871, China  
E-mail: gwyzlzsb@pku.edu.cn; x.yan@pku.edu.cn

Z. Zhang, R. Wang, C. Chang, J. Lou  
Innovation Laboratory of Terahertz Biophysics  
National Innovation Institute of Defense Technology  
Beijing 100071, China  
E-mail: loujing9486@163.com

Z. Wang  
State Key Laboratory of Surface Physics and Key Laboratory of Micro and Nano Photonic Structures (Ministry of Education)  
Fudan University  
Shanghai 200433, China

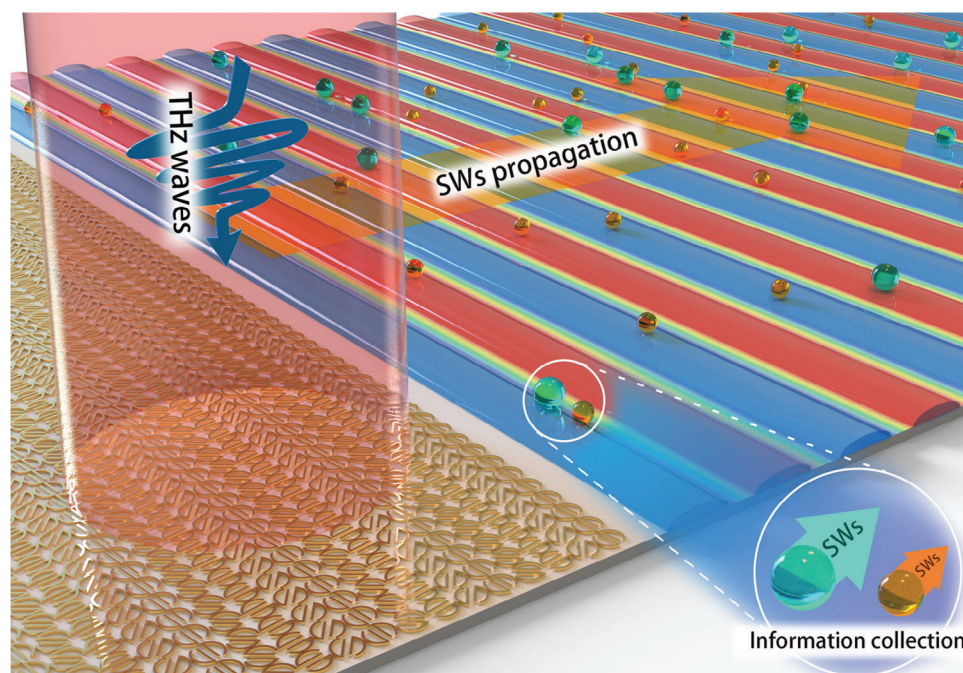
C. Zhang  
Department of Neurosurgery Center  
Zhujiang Hospital  
Southern Medical University  
Guangzhou 510282, China

Z. Yao, S. Zhang, Z. Tian, J. Han  
Center for Terahertz Waves and College of Precision Instrument and Optoelectronics Engineering  
Key Laboratory of Optoelectronic Information Technology (Ministry of Education of China)  
Tianjin University  
Tianjin 300072, China

C. Qiu  
Department of Electrical and Computer Engineering  
National University of Singapore  
Singapore 117583, Singapore  
E-mail: chengwei.qiu@nus.edu.sg

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adma.202308453>

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**Figure 1.** Schematic diagram of refractive sensing and fingerprint recognition using SWs metadvice. The incident waves directly irradiate the atomic structure of the PB meta-atoms and excites SWs to propagate on the right surface of the metadvice. The analytes information collected in the propagation is reflected by analysing the transmission characteristics of SWs.

continues to rise, a new generation of optical sensors has emerged. These sensors rely on the optical changes induced by bio-molecules to achieve highly sensitive detection, which are widely applicable due to their easily-use, low-cost, and strong universality.<sup>[7,8,11–14]</sup>

The operating frequency of an optical sensor is crucially linked to the vibration frequency of the bio-molecules. Many bio-molecules and tissues exhibit unique biological fingerprints within the terahertz (THz) range. The characterizes of this frequency range such as low energy, high penetration, and refined resolution, have propelled THz technology into a powerful tool for biological research. It is now extensively employed in organic molecular and biological studies that depend on resonance responses with fingerprint spectra. For instance, THz technology has recently been applied in investigating the ordered water molecule structure on neuron surfaces and studying the structure of biological macromolecules such as cell ion channel proteins.<sup>[15,16]</sup> Furthermore, THz technology has found extensive applications in bio-sensing and detection. It is utilized for tasks such as characterizing cell components and differentiating cancerous tissues, offering unique contributions to the fields of biological and medical research.<sup>[17–26]</sup>

Metasurfaces with subwavelength structures and field-enhanced properties are sensitive in identifying environmental changes in the fields of biological and chemical analyzing.<sup>[27–38]</sup> Numerous approaches have been proposed recently to utilize these properties for achieving highly sensitive sensing in 2D two-dimensional regions. Such as detecting minor changes of medium through near field enhancement,<sup>[12,19,34,38–41]</sup> high quality factor (Q-factor) metasurface sensing based on fingerprint spectra,<sup>[33,35,39]</sup> and improving resonance strength by

coupling magnetic dipoles with electric quadrupoles to enhance sensitivity.<sup>[36,42,43]</sup> However, traditional metasurface sensors typically rely on periodic structure with single resonant mode for acquiring and accumulating analytes information.<sup>[44–47]</sup> This approach relies on transmission or reflection spectra measurements, which decides the way of enhancing the interaction between analytes and light can only be achieved in a 2D structure and limits the ability to achieve higher sensitivity. Additionally, the subwavelength characteristics of metasurfaces and the demands for compact integration impose higher requirements on sample processing.<sup>[12,48]</sup> Therefore, finding a way to achieve greater acquisition and accumulation of information about the detected object remains a challenging problem.

Surface waves (SWs) are electromagnetic (EM) wave modes confined at the interface between metal and medium, resulting from the coupling of the EM waves with the surface plasmons, which are extensively utilized due to confined local fields and slow group velocities, such as antennas,<sup>[49,50]</sup> rejection filter,<sup>[51]</sup> graded-index lenses,<sup>[52]</sup> and so on. However, researchers have primarily focused on constructing well-designed metasurfaces to achieve complex manipulation of SWs wavefronts rather than sensitive sensing.<sup>[31,32,53–55]</sup> The long-range propagation capability of SWs has been overlooked, leading to the underutilization of the excellent sensing potential that SWs offer. Therefore, we expect to deeply delve the potential of SWs in the bio-sensing field.

In this paper, the long-range propagation property of SWs is utilized effectively as illustrated in **Figure 1**, enabling refractive sensing and fingerprint recognition. A metadvice based on Pancharatnam–Berry (PB) meta-atoms is designed and experimentally characterized for the excitation and propagation of SWs. Our results exhibit sensitive sensing in both transmission

phase and amplitude, corresponding to refractive sensing and fingerprint recognition, respectively. The refractive sensing performance exhibits the sensitivity up to 215.5°/RIU and indicates around fourfold substantial phase-shift compared to traditional transmission-type metasurface.<sup>[1,56]</sup> Moreover, within 5 μm thickness of polyimide, quartz and silicon nitride layers, the propagation phase difference of 91.1°, 101.8°, and 126.4° is experimentally obtained, respectively. Besides, the capability to simultaneously detect multiple fingerprint spectra within a continuous spectrum is validated using compact SWs metadvice, eliminating additional array arrangements for individual frequency matching.<sup>[8,57–61]</sup> The performance comparison of the reported metasurface sensors with our strategy could be seen in Table S1 (Supporting Information). This work establishes a novel paradigm in the realm of metasurface sensing, offering a wide array of potential applications in bio-sensing and other fields.

## 2. Results and Discussions

### 2.1. Design and Characterization of the SWs Metadevices

The design of PB meta-atoms is first considered, and then the SWs metadvice is further constructed. Consider the meta-atoms exhibiting optical responses to linearly polarized (LP) waves along the *x*-polarized (*x*-pol) and *y*-polarized (*y*-pol), which can be described by Jones matrices

$$J = \begin{bmatrix} r_{xx} & 0 \\ 0 & r_{yy} \end{bmatrix} \quad (1)$$

where the  $r_{xx} = R_{xx}e^{i\varphi_{xx}}$  and  $r_{yy} = R_{yy}e^{i\varphi_{yy}}$  represent the reflection coefficient of *x*-pol and *y*-pol incident, respectively. For the metadvice composed of PB unit atoms, circularly polarized (CP) waves incidence with definite handedness always result in two reflective waves, i.e., a normal and an anomalous waves.<sup>[31,62,63]</sup> The normal wave corresponds to mirror reflection of the incident EM waves, while the anomalous wave results from the mutual interference of scattered light from PB meta-atoms. The efficiencies of the normal and anomalous waves are  $R_n = |(r_{xx} + r_{yy})|^2/4$  and  $R_a = |(r_{xx} - r_{yy})|^2/4$ , respectively. Therefore, the coupling efficiency of SWs in the device can be described by the efficiency of extraordinary waves.<sup>[62]</sup> Without considering the dielectric loss of the device, the maximum efficiency condition for extraordinary waves can be further obtained<sup>[63]</sup>

$$|r_{xx}| = |r_{yy}| = 1, \Delta\varphi = \varphi_{xx} - \varphi_{yy} = \pm\pi \quad (2)$$

This condition implies that the PB meta-atoms should play a role as reflecting half plates under maximum efficiency. To convert the incident PWs into SWs transmitted along a 2D plane with high coupling efficiency, a metasurface with linear phase delay must be constructed through periodic arrangement of PB meta-atoms. This is achieved by meeting the condition  $\varphi(x, y) = \varphi_0 + \xi_x x$  ( $\xi_x > k_0$ ), where  $\xi_x$  represents the phase gradient of adjacent PB meta-atoms periodically arranged in the *x*-direction and  $k_0$  represents free space wavevector.<sup>[64]</sup>

Here, a classical metal-insulator-metal cavity is employed to design the structure of PB meta-atoms, as illustrated in Figure 2a.

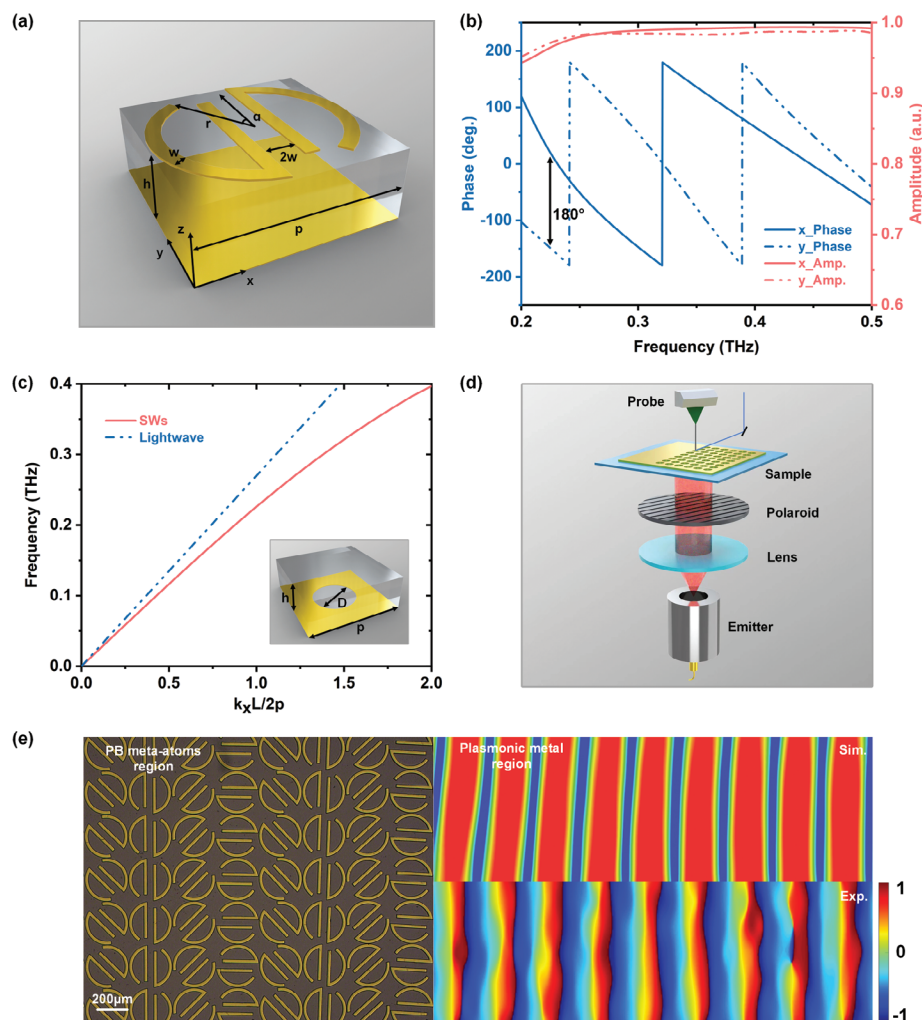
The PB meta-atom comprise a resonator antenna made of a gold ring (Au,  $\sigma_{Au} = 4.56e7 \text{ S m}^{-1}$ ) connected with gold rods and reflector, with quartz (SiO<sub>2</sub>,  $\epsilon_r = 3.75 + 0.001i$ ) serving as the substrate material for separation purposes. The parameters of the PB meta-atom include period  $p = 277.6 \text{ μm}$ , resonant ring width  $w = 20 \text{ μm}$ , radius  $r = 126.6 \text{ μm}$ , open-angle  $\alpha = 33^\circ$  and quartz substrate thickness  $h = 100 \text{ μm}$ . The reflection amplitude and phase of PB meta-atoms when irradiated by the *x*-pol and *y*-pol incident waves are simulated using commercial software CST Microwave Studio 2022, as depicted in Figure 2b. By controlling the thickness of the quartz substrate, magnetic resonance occurs under the *x*-pol and *y*-pol incidence in an underdamped region. The results demonstrate that PB meta-atoms exhibit high reflectivity across the entire frequency band. Additionally, the phase shift covers nearly the full range of  $2\pi$ , with the reflected phase difference between the *x*-pol and *y*-pol being close to  $\pi$  throughout the frequency band. This indicates that our device can effectively operate around 0.21–0.26 THz with a relative bandwidth of 21.3%. To achieve efficient excitation and regular wavefront of SWs, the operating frequency of the device is set at 0.225 THz.

The SWs driven by PWs flow confined within the PB meta-atoms region and can be guided to other systems efficiently under the condition of wavevector matching. In order to utilize SWs more efficiently, a plasmonic metal is designed to support the propagation of SWs. The plasmonic metal structure is depicted in the illustration in Figure 2c, comprising a quartz substrate and a gold reflector with an air hole, where the air hole diameter  $D = 138.8 \text{ μm}$ . Simulated results of the dispersion relation of the plasmonic metals are displayed in Figure 2c, where the center frequency is matched with 0.225 THz.

Next, the designed PB meta-atoms are integrated with the plasmonic metal to construct the SWs metadvice. The micrograph on the left in Figure 2e demonstrates the arrangement of identical PB meta-atoms periodically along the *x*-direction with a specific rotation angle  $\Delta\varphi = 45^\circ$ . Consequently, based on the PB mechanism,<sup>[62]</sup> the meta-atoms possess a particular spin- $\sigma$  (where  $\sigma = +$  denotes left-handed circular polarization, and  $\sigma = -$  indicates right-handed circular polarization) of CP incidence. When the light irradiate on PB meta-atoms with such an arrangement, it will exhibit the following phase distribution

$$\Phi^\sigma = \Phi_0 + \sigma\xi \cdot x \quad (3)$$

The lattice constant of the arranged PB meta-atoms super cell is  $L = 4p = 1110.4 \text{ μm}$ , resulting in a phase gradient of  $\xi = 1.2k_0$ . This phase gradient enables a complete  $2\pi$  phase change within four adjacent PB meta-atoms arranged at a specific rotation angle. Finally, connecting the arranged PB meta-atoms with the plasmonic metal yields the metadvice with the capability of exciting and propagating SWs under normal CP incidence at 0.225 THz. Notably, the SWs coupling scheme does not require strict wavevector matching with plasmonic metal,<sup>[64]</sup> which allows the metadvice having high coupling efficiency from PWs to SWs in the operating band around 0.225 THz (Note S2, Supporting Information). Figure 2e presents the validation of SWs excited by metadvice through simulated and experimental results, where the left and right sides represent the PB meta-atoms region and plasmonic metal region of the metadvice, respectively. The upper right side of Figure 2e displays the simulated  $\text{Re}(E_z)$  field



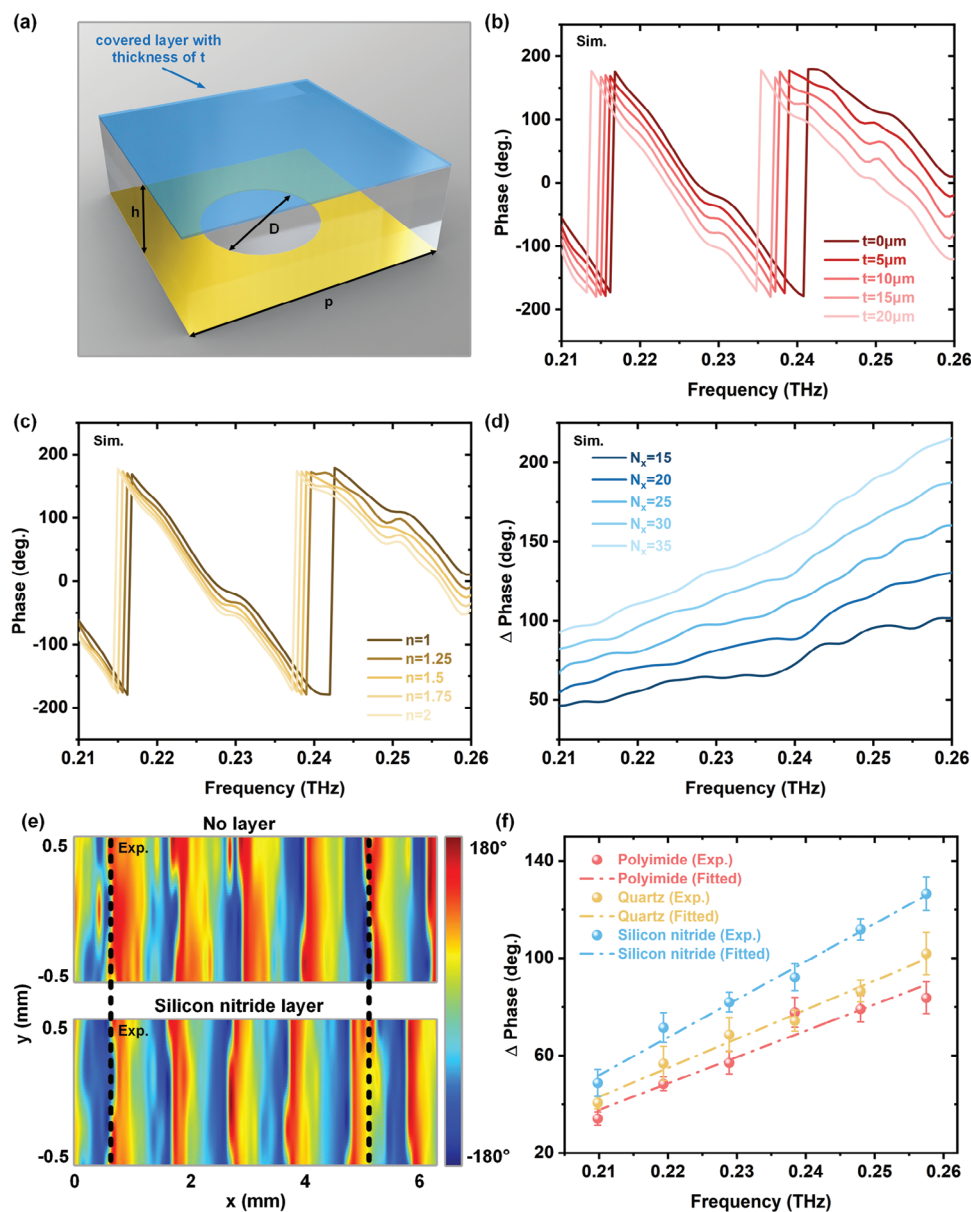
**Figure 2.** a) Schematic diagram of the designed PB meta-atom structure, where  $p = 277.6 \mu m$ ,  $w = 20 \mu m$ ,  $r = 126.6 \mu m$ ,  $\alpha = 33^\circ$ , and  $h = 100 \mu m$ . b) Simulated reflection spectra of the amplitude and phase of the PB meta-atoms under  $x$ -pol and  $y$ -pol incidences. c) Simulated dispersion relation of SWs supported by the plasmonic metal. The inset shows the schematic structure of the plasmonic metal unit cell, where  $D = 138.9 \mu m$ . d) Schematic diagram of the THz near-field scanning system used for characterizing SWs. e) Simulated and experimental fitting of SWs metadvice. The two sides of the figure represent the PB meta-atoms region and plasmonic metal region, respectively. The left side displays the micrograph of PB meta-atoms array; the upper right side presents the simulated field distributions of  $Re(E_z)$  at 0.225 THz, and the lower right side shows the measured field distributions of  $Re(E_z)$  at 0.225 THz.

distributions when the coupled SWs propagate along the plasmonic metal surface at 0.225 THz, which reveals a regular wavefront and a favorable propagation effect. To characterize the performance of the excited SWs, the THz near-field scanning system (the experimental device schematic is shown in Figure 2d) is utilized (Note S5, Supporting Information). During the measurement, the air holes drilled beneath the plasmonic metal allow the excited SWs leaking from the sample. The diameter  $D$  is determined by experimental requirements, where the impact of  $D$  on the dispersion relation could be negligible. The probe, positioned around  $75 \mu m$  away from the sample on a movable plane, facilitates the collection of local electric field information. The measured field distribution of  $Re(E_z)$  of SWs at 0.225 THz is depicted in the lower right side Figure 2e. It is evident that SWs with regular waveforms exists on the plasmonic metal surface, and the results align well with the simulations. Notably, both in

simulations and experiments, SWs can achieve long-range propagation on the plasmonic metal surface and accumulate a substantial amount of spatial information and phase changes during propagation. This unique property offers a novel approach for enhancing the interaction between EM waves and matter within the metadvice, which forms the basis for realizing highly sensitive bio-molecular sensing.

## 2.2. Refractive Sensing and Fingerprint Recognition based on SWs Meta-devices

In the previous section, the experimental fitting of the metadvice to the performance of excited SWs has been verified. To further substantiate the continuous information accumulation capabilities of SWs, simulations are performed using classical resonator



**Figure 3.** a) Schematic representation of a plasmonic metal surface covered with an analyte layer. b) The plasmonic metal length is set as  $N_x$  times the cell period, indicating the transmission distance. Under the condition of  $N_x = 20$ , the simulated transmission phase of SWs for various thicknesses  $t$  of analyte layers ( $n = 2$ ). c) Analysis of the variation in SWs transmission phase with different  $n$  of the analyte layer at  $t = 10 \mu\text{m}$  and  $N_x = 20$ . d) Investigation of differences in transmission phase at  $t = 10 \mu\text{m}$  and various propagation distances  $N_x$  for two analyte layers with  $n = 1$  and  $n = 2$ , respectively. e) The measured phase distributions of SWs at 0.25 THz, with the condition of without analyte layer and covered with a silicon nitride layer with thickness  $t = 5 \mu\text{m}$ , respectively. f) When the propagation distance is set as  $N_x = 30$ , the disparities in SWs propagation phase between various analyte layers and without analyte layer are measured and linearly fitted. Error bars are derived from three distinct sets of experimental data.

units to verify the sensing performance of the metadvice.<sup>[51]</sup> The results demonstrate that various analytes exert distinct influences on SWs, and this influence is consistently maintained throughout the entire propagation process. This phenomenon reflects the information accumulation ability of SWs based on long-range propagation (Note S3, Supporting Information).

To further investigate the sensing performance and the ability to detect fingerprint spectra of the metadvice, the propagation information of SWs is examined using analyte layers and molecular layers with predetermined dielectric constants. In this

section, we first assume that an analyte layer with certain refractive index  $n$  is uniformly distributed across the entire plasmonic metal surface, as depicted in **Figure 3a**. Comprehensive information about the analyte layers is obtained by controlling the thickness within the range of the decay length of SWs in air. The transmission distance of SWs on the metadvice surface allows us to verify the capability of accumulating information from both the propagation phase and amplitude. Here, propagation distance is set as the plasmonic metal length, which being  $N_x$  times to the cell period.  $N_x = 20$  and refractive index  $n = 2$  of analyte layers

are selected as the conditions to investigate the effect of thickness  $t$  on SWs propagation. We fix the initial phase of SWs and observe the transmission phase at the tail of plasmonic metal. Figure 3b shows that the transmission phase is uniformly influenced by the analyte layers across the entire operating frequency and exhibits a regular change with increasing thickness  $t$ . The uniform influence arises from the real and null imaginary parts of the refractive index, which corresponds to the absence of distinct resonance modes within the operating frequency. As a result, the impact of the analyte layers is revealed through the overall deviation in the dispersion relation of the plasmonic metal, and the alteration in the equivalent wavevector leads to a shift in the propagation phase. Thicker analyte layers correspond to obvious phase changes, which indicates the analytes content could be reflected by the propagation phase difference.

For the conventional sensing strategy based on metasurfaces, the analyte information is typically analyzed and distinguished based on transmission and reflection spectra. Although the interaction between EM waves and the analytes is continually enhanced due to the compact and localized character of metasurface. However, when the measured information is integrated in the transmission and reflection spectra, the phase information of the EM wave is lost.

Additionally, by setting  $t = 10 \mu\text{m}$  and  $N_x = 20$ , we analyze the influence of the refractive index on the SWs transmission. As shown in Figure 3c, the refractive index change of the analyte layers influences the equivalent wavevector and be reflected by the transmission phase variation of SWs, which aligns with the discussion on thickness. However, the long-range propagation character of SWs is not fully reflected. Considering the spatial interaction and information accumulation of analytes, we further discuss the influence of propagation distance  $N_x$ . At  $t = 10 \mu\text{m}$ , analyte layers with refractive index of  $n = 1$  and  $n = 2$  are covered with plasmonic metal respectively, and the differences in propagation phase between  $n = 1$  and  $n = 2$  are compared with changes in  $N_x$ . As the propagation distance increases, the propagation phase difference of SWs also increases under the coverage of analyte layers with  $n = 1$  and  $n = 2$ , respectively, as illustrated in Figure 3d. The pattern of information accumulation and a linear increase tendency can be observed (Note S4, Supporting Information). Furthermore, within the operating frequency, higher frequencies correspond to more evident variation in the dispersion relation of plasmonic metal and exhibit more evident phase difference and more sensitive sensing performance. For the common metasurface sensors that rely on transmission and reflection spectra, the interaction of EM waves with narrow and thin metasurfaces is severely limited. However, by exploiting the propagation properties of SWs, sensitive refractive sensing is achieved, which is exemplified by the  $215.5^\circ/\text{RIU}$  at  $N_x = 35$  and can be further enhanced by the propagation length. It is worth emphasizing that the analyte layer thickness discussed in Figure 3d does not reach the maximum value. In the actual experiments, the limitation imposed by computing performance can be disregarded to achieve longer propagation distances, and the sensing and detection performance of SWs could be further improved.

To enhance the evidential support for the sensing application of the metadvice, further experiments with real analyte layers are conducted. We examine polyimide ( $n_{\text{polyimide}} = 1.8$ ), quartz ( $n_{\text{quartz}} = 1.95$ ), and silicon nitride ( $n_{\text{silicon nitride}} = 2.7$ ) layers, each

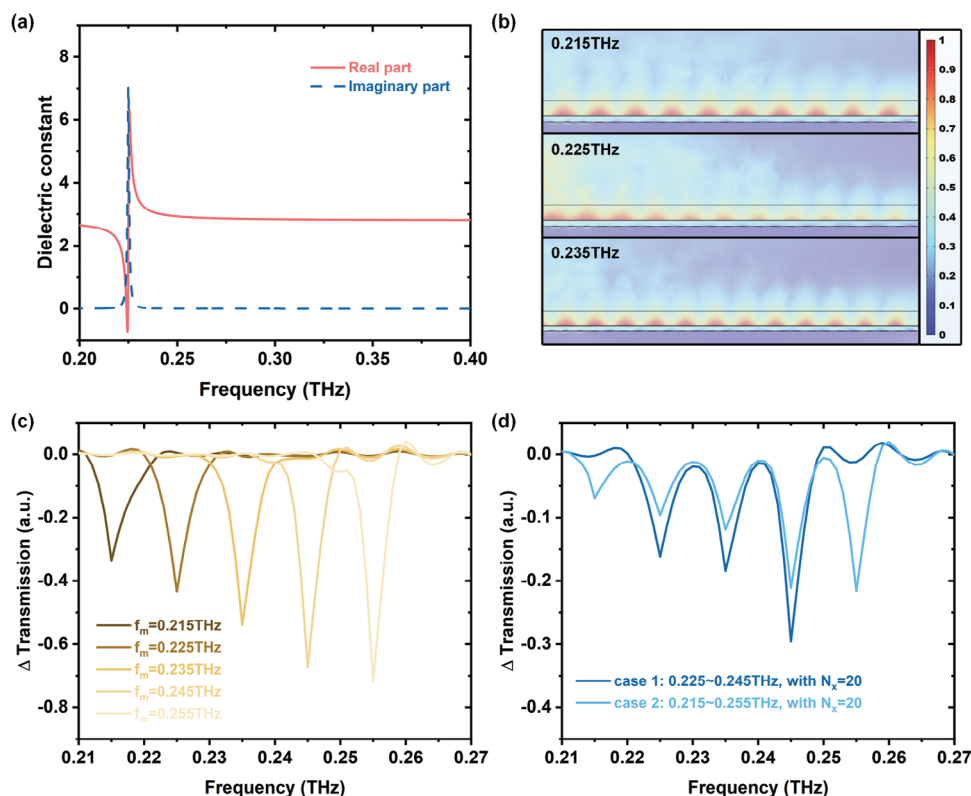
characterized by a stable refractive index and a low absorption coefficient around operating frequency.<sup>[65–68]</sup> These analyte layers provide a suitable platform to investigate the sensing performance of the metadvice. The plasmonic metal surface is uniformly covered by the analyte layers with thickness around  $t = 5 \mu\text{m}$ , where adopts the rotational deposition and the chemical vapor deposition (CVD) methods, respectively (Note S6, Supporting Information). As depicted in Figure 3e, the phase distribution of SWs at 0.25 THz is experimentally measured without analyte layer and with the silicon nitride layer ( $t = 5 \mu\text{m}$ ), respectively. The impact of silicon nitride on the dielectric environment manifests in an overall shift of the dispersion relation of plasmonic metal, resulting in an increase of the equivalent wavevector. Consequently, within the same propagation distance (indicated by the black dashed lines), SWs exhibit a shorter phase period when the plasmonic metal surface is covered with the sensing analytes.

With the increase of the propagation distance  $N_x$ , the influence of the dielectric environment is accumulated continuously in the propagation phase of SWs, corresponding to more significant phase difference. Under the same propagation distance of  $N_x = 30$ , the propagation phase differences induced by the analyte layers are experimentally measured and fitted, as described in Figure 3f. It is evident that covering analyte layers result in a faster phase change of SWs compared with the condition of bare device, and the phase differences is enhanced with the increase of frequency and refractive index. These results exhibit a well-fitted linear relationship, which are consistent with the simulations (Figure 3d). When the plasmonic metal is covered with  $5 \mu\text{m}$  thickness of polyimide, quartz and silicon nitride layers, the maximum phase change of  $91.1^\circ$ ,  $101.8^\circ$ , and  $126.4^\circ$  is achieved, respectively, which is unattainable by other works.

Considering the fingerprint spectra of many bio-molecules are distributed in the THz band, our aim is to explore the ability of the metadvice to recognize fingerprint spectra and expand its application in the field of bio-molecular sensing. To achieve this, further studies are conducted by incorporating the Drude–Lorentz model. Given the significance of the Drude–Lorentz model for describing materials, we employ the commercial software COMSOL Multiphysics 6.1 to simulate the behavior of the molecular layers. Supposing a thin layer of molecules distribute on the plasmonic metal surface with same manner, and the dielectric constant can be appropriately described using the Drude–Lorentz model<sup>[69–72]</sup>

$$\epsilon_m(\omega) = \epsilon_\infty + \frac{\omega_p^2}{\omega_m^2 - \omega^2 - i\Gamma_m\omega} \quad (4)$$

where  $\epsilon_\infty$  is the background permittivity,  $\omega_p$  is the plasma frequency,  $\omega_m$  is the vibrational resonant frequency, and  $\Gamma_m$  is the vibrational damping rate. By adjusting these parameters, the vibration resonant frequency  $f_m$  of the molecular layers can be positioned within the operating frequency range, as depicted in Figure 4a. This figure illustrates the effective dielectric constant curve of the molecular layer at  $f_m = 0.225 \text{ THz}$ . The information accumulation process of SWs for molecules with specific vibration modes is simulated, by applying molecular layers onto the metadvice surface using the same settings as the analyte layers, with  $t = 10 \mu\text{m}$  and  $N_x = 20$ . As illustrated in Figure 4b, the  $\text{Re}(E_z)$  field distributions of SWs shows a distinct vibration



**Figure 4.** a) The covered layer is modelled with the Drude–Lorentz mode, with the vibration frequency  $f_m = 0.225$  THz and the background permittivity is  $\epsilon_\infty = 2.8$ . b) Within a molecular layer thickness of  $t = 10$   $\mu\text{m}$  and vibration frequency of  $f_m = 0.225$  THz, the  $\text{Re}(E_z)$  field distributions of transformed SWs are simulated, when EM waves with different frequencies are irradiated onto the metadvice. c) The normalized amplitude differences are obtained when the covered layer is set as single-frequency vibration of 0.215, 0.225, 0.235, 0.245, 0.255 THz, respectively, compared with the molecular layer with  $\epsilon_\infty = 2.8$ . d) The normalized amplitude differences are obtained when the covered layer is set as two multifrequency vibration cases, compared with the molecular layer with  $\epsilon_\infty = 2.8$ .

mode within the frequency range around  $f_m = 0.225$  THz, as predicted by the Drude–Lorentz model. The sharp virtual part of the effective dielectric constant induces resonance effects near the vibration mode's frequency, leading to significant absorption and amplitude attenuation. This phenomenon enables an effective method to simulate bio-molecules with specific fingerprint. Remarkably, the metadvice allows for the excitation of SWs across the entire power frequency range, offering the capability to tune SWs frequencies over a wide spectrum. Consequently, when the SWs operating frequency aligns with the molecular fingerprint spectra, the fingerprint information becomes directly evident in the transmission amplitude of SWs.

Molecular layers with Drude–Lorentz model and different  $f_m$  are further compared with the condition of background dielectric constant  $\epsilon_\infty = 2.8$ , where simulate the scenarios akin to bio-molecular solutions and pure solution media. In Figure 4c, the normalized decreases in the transmission amplitude as the molecular layers with varying  $f_m$  propagate compared to  $\epsilon_\infty = 2.8$  are displayed. By adjusting the  $f_m$  within the operating frequency, the vibration modes of the molecular layer become distinctly evident in the decline of the transmission curve. The extent of this decline relates to the interaction between SWs and the molecular layer, offering the content information of the molecules to a certain degree. By combining different  $f_m$  values with their respective full width at half maximum (FWHM), the average Q-factor

( $f_m/\text{FWHM}$ ) is calculated to be 74, which exceeds the typical values of metallic antenna.<sup>[73,74]</sup>

To assess the metadvice capability of simultaneously detecting multiple fingerprint spectra, two cases are set up as following. In both cases,  $t = 10$   $\mu\text{m}$  and  $N_x = 20$  are maintained. In case 1, three different molecular layers with  $f_m$  values of 0.225, 0.235, and 0.245 THz are alternately arranged simultaneously, with the alternating interval of  $N_x = 2$ . In case 2, five different molecular layers with  $f_m$  values of 0.215, 0.225, 0.235, 0.245, and 0.255 THz are also arranged, similar to case 1. Figure 4d presents the transmission amplitude decline curve, which clearly reflects the vibration modes of all molecular layers. Remarkably, different vibration modes only affect a small portion of the SWs transmission within the relevant frequency range, enabling the accumulation of various vibration mode information without mutual interference during propagation. In case 2, the reduced content of each molecular layer leads to a corresponding lower amplitude reduction due to the relatively lower content. Nonetheless, this observation validates the ability to discern the content of molecules being detected. Unlike existed studies, our metadvice directly reflects the frequency position of the fingerprint spectra from the transmission curve, rather than submerging the frequency information in the whole transmission curve.<sup>[17,18,48]</sup> Furthermore, our metadvice achieves continuous fingerprint spectra acquisition through compact meta-atoms structure, avoiding

multiple meta-atom structures to match individual frequencies within the band.<sup>[57]</sup> This substantially broadens the sensing application range and reduces structural complexity. As a result, our metadvice finds versatile applications in diverse bio-sensing and specific detection conditions.

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### 3. Conclusions

In conclusion, we propose a novel strategy for achieving refractive sensing and fingerprint recognition using SWs. The capability of the metadvice is experimentally validated to excite and propagate SWs effectively over a long range within THz frequency range. Furthermore, the metadvice effectively represents information in the transmission phase of SWs for analyte layers with specific refractive index, exhibiting sensitivity values of 215.5°/RIU, which surpass traditional metasurface sensing methods. Under the condition of 5 μm thickness of polyimide, quartz and silicon nitride layers, the maximum propagation phase difference of 91.1°, 101.8°, and 126.4° is experimentally achieved, respectively. Additionally, we successfully achieve specific fingerprint recognition based on the transmission amplitude of SWs within a continuous spectrum, eliminating complex structures of pixelated metasurfaces. Overall, our work introduces a novel paradigm for common metasurface sensing, exploring fresh methods of enhancing the interaction between EM waves and target analytes, while exploiting the application potential of metasurface-excited SWs. These findings hold crucial value for future applications in refractive sensing and fingerprint recognition fields.

### Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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### Conflict of Interest

The authors declare no conflict of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

### Keywords

fingerprint recognition, metasurface, plasmonic coupler, refractive sensing, surface waves

- [1] J. Lou, Y. Jiao, R. Yang, Y. Huang, X. Xu, L. Zhang, Z. Ma, Y. Yu, W. Peng, Y. Yuan, Y. Zhong, S. Li, Y. Yan, F. Zhang, J. Liang, X. Du, C. Chang, C.-W. Qiu, *Proc. Natl. Acad. Sci. USA* **2022**, *119*, 2209218119.
- [2] S. Li, Q. Li, W. Chen, Z. Song, Y. An, P. Chen, Y. Wu, G. Wang, Y. He, Q. Miao, *Small* **2022**, *18*, 2201334.
- [3] J. Kim, A. S. Campbell, B. E.-F. De Ávila, J. Wang, *Nat. Biotechnol.* **2019**, *37*, 389.
- [4] J. Chen, G. Oudeng, H. Feng, S. Liu, H.-W. Li, Y.-P. Ho, Y. Chen, Y. Tan, M. Yang, *Small* **2022**, *18*, 2201779.
- [5] J. Li, Y. Liu, L. Yuan, B. Zhang, E. S. Bishop, K. Wang, J. Tang, Y.-Q. Zheng, W. Xu, S. Niu, L. Beker, T. L. Li, G. Chen, M. Diyaolu, A.-L. Thomas, V. Mottini, J. B.-H. Tok, J. C. Y. Dunn, B. Cui, S. P. Pasca, Y. Cui, A. Habtezion, X. Chen, Z. Bao, *Nature* **2022**, *606*, 94.
- [6] F. Meng, W. Yu, C. Chen, S. Guo, X. Tian, Y. Miao, L. Ma, X. Zhang, Y. Yu, L. Huang, K. Qian, J. Wang, *Small* **2022**, *18*, 2200784.
- [7] M. D. Baaske, M. R. Foreman, F. Vollmer, *Nat. Nanotechnol.* **2014**, *9*, 933.
- [8] F. Yesilkoy, E. R. Arvelo, Y. Jahani, M. Liu, A. Tittl, V. Cevher, Y. Kivshar, H. Altug, *Nat. Photonics* **2019**, *13*, 390.
- [9] X. Mei, H.-C. Lee, K.-Y. Diao, M. Huang, B. Lin, C. Liu, Z. Xie, Y. Ma, P. M. Robson, M. Chung, A. Bernheim, V. Mani, C. Calcagno, K. Li, S. Li, H. Shan, J. Lv, T. Zhao, J. Xia, Q. Long, S. Steinberger, A. Jacobi, T. Deyer, M. Lukasz, F. Liu, B. P. Little, Z. A. Fayad, Y. Yang, *Nat. Med.* **2020**, *26*, 1224.
- [10] T. Chaibun, J. Puenpa, T. Ngamdee, N. Boonapatcharoen, P. Athamanolap, A. P. O'mullane, S. Vongpunsawad, Y. Poovorawan, S. Y. Lee, B. Lertanantawong, *Nat. Commun.* **2021**, *12*, 802.
- [11] S. Sakamoto, W. Putalun, S. Vimolmangkang, W. Phoolcharoen, Y. Shoyama, H. Tanaka, S. Morimoto, *J. Nat. Med.* **2018**, *72*, 32.
- [12] L. Xie, W. Gao, J. Shu, Y. Ying, J. Kono, *Sci. Rep.* **2015**, *5*, 8671.
- [13] M. U. Ahmed, I. Saaem, P. C. Wu, A. S. Brown, *Crit. Rev. Biotechnol.* **2014**, *34*, 180.
- [14] S. Shrivastava, T. Q. Trung, N.-E. Lee, *Chem. Soc. Rev.* **2020**, *49*, 1812.
- [15] Z. Xiang, C. Tang, C. Chang, G. Liu, *Kexue Tongbao (Foreign Lang. Ed.)* **2020**, *65*, 308.
- [16] Z. Zhu, J. Zhang, Y. Song, C. Chang, G. Ren, J. Shen, Z. Zhang, T. Ji, M. Chen, H. Zhao, *Analyst* **2020**, *145*, 6006.
- [17] L. Yu, L. Hao, T. Meiqiong, H. Jiaoqi, L. Wei, D. Jinying, C. Xueping, F. Weiling, Z. Yang, *RSC Adv.* **2019**, *9*, 9354.
- [18] X. Yang, K. Yang, Y. Luo, W. Fu, *Appl. Microbiol. Biotechnol.* **2016**, *100*, 5289.
- [19] M. Beruete, I. Jáuregui-López, *Adv. Opt. Mater.* **2019**, *8*, 1900721.
- [20] J. F. O'hara, W. Withayachumnankul, I. Al-Naib, *J. Infrared, Millimeter, and Terahertz Waves* **2012**, *33*, 245.
- [21] W. Peng, Z. Zhu, J. Lou, K. Chen, Y. Wu, C. Chang, *eLight* **2023**, *3*, 18.
- [22] A. Belushkin, F. Yesilkoy, J. J. González-López, J. C. Ruiz-Rodríguez, R. Ferrer, A. Fàbrega, H. Altug, *Small* **2020**, *16*, 1906108.
- [23] D. Etezadi, J. B. Warner, H. A. Lashuel, H. Altug, *ACS Sens.* **2018**, *3*, 1109.
- [24] Y. Jahani, E. R. Arvelo, F. Yesilkoy, K. Koshelev, C. Cianciaruso, M. De Palma, Y. Kivshar, H. Altug, *Nat. Commun.* **2021**, *12*, 3246.
- [25] D. Yoo, D. A. Mohr, F. Vidal-Codina, A. John-Herpin, M. Jo, S. Kim, J. Matson, J. D. Caldwell, H. Jeon, N.-C. Nguyen, L. Martin-Moreno, J. Peraire, H. Altug, S.-H. Oh, *Nano Lett.* **2018**, *18*, 1930.
- [26] D. Kavungal, P. Magalhães, S. T. Kumar, R. Kolla, H. A. Lashuel, H. Altug, *Sci. Adv.* **2023**, *9*, eadg9644.
- [27] N. Liu, H. Guo, L. Fu, S. Kaiser, H. Schweizer, H. Giessen, *Nat. Mater.* **2008**, *7*, 31.

- [28] N. I. Zheludev, Y. S. Kivshar, *Nat. Mater.* **2012**, *11*, 917.
- [29] A. V. Kildishev, A. Boltasseva, V. M. Shalae, *Science* **2013**, *339*, 1232009.
- [30] H. Zang, Z. Xi, Z. Zhang, Y. Lu, P. Wang, *Sci. Adv.* **2022**, *8*, eadd1973.
- [31] Z. Wang, S. Li, X. Zhang, X. Feng, Q. Wang, J. Han, Q. He, W. Zhang, S. Sun, L. Zhou, *Adv. Sci. (Weinh)* **2020**, *7*, 2000982.
- [32] S. Li, Z. Wang, S. Dong, S. Yi, F. Guan, Y. Chen, H. Guo, Q. He, L. Zhou, S. Sun, *Nanophotonics* **2020**, *9*, 3473.
- [33] S. R. Raz, M. Bremer, W. Haasnoot, W. Norde, *Anal. Chem.* **2009**, *81*, 7743.
- [34] S. J. Park, J. T. Hong, S. J. Choi, H. S. Kim, W. K. Park, S. T. Han, J. Y. Park, S. Lee, D. S. Kim, Y. H. Ahn, *Sci. Rep.* **2014**, *4*, 4988.
- [35] F. Miyamaru, K. Hattori, K. Shiraga, S. Kawashima, S. Suga, T. Nishida, M. W. Takeda, Y. Ogawa, *J. Infrared, Millimeter, and Terahertz Waves* **2013**, *35*, 198.
- [36] B. Liu, Y. Peng, Z. Jin, X. Wu, H. Gu, D. Wei, Y. Zhu, S. Zhuang, *Chem. Eng. J.* **2023**, *462*, 142347.
- [37] R. Wang, L. Xu, L. Huang, X. Zhang, H. Ruan, X. Yang, J. Lou, C. Chang, X. Du, *Small* **2023**, *19*, 2301165.
- [38] Y. Jiao, J. Lou, Z. Ma, L. Cong, X. Xu, B. Zhang, D. Li, Y. Yu, W. Sun, Y. Yan, S. Hu, B. Liu, Y. Huang, L. Sun, R. Wang, R. Singh, Y. Fan, C. Chang, X. Du, *Mater. Horiz.* **2022**, *9*, 2984.
- [39] L. Hongkui, F. Yao, *Optik* **2023**, *276*, 170656.
- [40] P. Wang, J. Lou, Y. Yu, L. Sun, L. Sun, G. Fang, C. Chang, *Nano Res.* **2023**, *16*, 7304.
- [41] S. K. Patel, J. Surve, J. Parmar, *Diamond Relat. Mater.* **2022**, *129*, 109367.
- [42] A. Ahmadivand, B. Gerislioglu, P. Manickam, A. Kaushik, S. Bhansali, M. Nair, N. Pala, *ACS Sens.* **2017**, *2*, 1359.
- [43] R. Zhao, Y. Feng, H. Ling, X. Zou, M. Wang, G. Lu, *Sensors (Basel)* **2023**, *23*, 2496.
- [44] O. Kameshkov, V. Gerasimov, S. Kuznetsov, *Photonics* **2023**, *10*, 243.
- [45] X. Lu, H. Ge, Y. Jiang, Y. Zhang, *Biosensors (Basel)* **2022**, *12*, 471.
- [46] L. Cong, S. Tan, R. Yahiaoui, F. Yan, W. Zhang, R. Singh, *Appl. Phys. Lett.* **2015**, *106*, 031107.
- [47] X. Lu, H. Ge, Y. Jiang, Y. Zhang, *Front. Phys.* **2022**, *10*, 973033.
- [48] X. Hu, G. Xu, L. Wen, H. Wang, Y. Zhao, Y. Zhang, D. R. S. Cumming, Q. Chen, *Laser Photonics Rev.* **2016**, *10*, 962.
- [49] Y.-F. Cheng, X. Ding, F. Peng, J. Feng, C. Liao, *IEEE Trans. Antennas Propag.* **2020**, *68*, 6389.
- [50] C. Wang, E. Li, D. F. Sievenpiper, *IEEE Trans. Antennas Propag.* **2017**, *65*, 5052.
- [51] S. Zhao, H. C. Zhang, J. Zhao, W. X. Tang, *Sci. Rep.* **2017**, *7*, 10576.
- [52] J. Zhao, L.-Z. Yin, F.-Y. Han, Y.-D. Wang, T.-J. Huang, C.-H. Du, P.-K. Liu, *Opt. Express* **2021**, *29*, 36366.
- [53] D. Wang, K. Liu, X. Li, G. Wang, S. Tang, T. Cai, *Nanophotonics* **2022**, *11*, 1177.
- [54] M.-Z. Chong, Y. He, J. Zhao, Y.-Y. Zhang, Z.-K. Zhang, C.-Q. Zhang, C.-H. Du, X. Zang, P.-K. Liu, *Nanoscale* **2023**, *15*, 4515.
- [55] S. Xiao, F. Zhong, H. Liu, S. Zhu, J. Li, *Nat. Commun.* **2015**, *6*, 8360.
- [56] Y. Li, Z. Deng, C. Qin, S. Wan, B. Lv, C. Guan, J. Yang, S. Zhang, J. Shi, *Opt. Express* **2023**, *31*, 492.
- [57] A. Tittl, A. Leitis, M. Liu, F. Yesilkoy, D.-Y. Choi, D. N. Neshev, Y. S. Kivshar, H. Altug, *Science* **2018**, *360*, 1105.
- [58] A. Leitis, A. Tittl, M. Liu, B. H. Lee, M. B. Gu, Y. S. Kivshar, H. Altug, *Sci. Adv.* **2019**, *5*, eaaw2871.
- [59] J. Lyu, S. Shen, L. Chen, Y. Zhu, S. Zhuang, *Photonix* **2023**, *4*, 28.
- [60] A. John-Herpin, D. Kavungal, L. Von Mücke, H. Altug, *Adv. Mater.* **2021**, *33*, 2006054.
- [61] D. Rodrigo, A. Tittl, N. Ait-Bouziad, A. John-Herpin, O. Limaj, C. Kelly, D. Yoo, N. J. Wittenberg, S.-H. Oh, H. A. Lashuel, H. Altug, *Nat. Commun.* **2018**, *9*, 2160.
- [62] W. Luo, S. Sun, H.-X. Xu, Q. He, L. Zhou, *Phys. Rev. Appl.* **2017**, *7*, 044033.
- [63] W. Luo, S. Xiao, Q. He, S. Sun, L. Zhou, *Adv. Opt. Mater.* **2015**, *3*, 1102.
- [64] S. Sun, Q. He, S. Xiao, Q. Xu, X. Li, L. Zhou, *Nat. Mater.* **2012**, *11*, 426.
- [65] T. Cai, B. Zheng, J. Lou, L. Shen, Y. Yang, S. Tang, E. Li, C. Qian, H. Chen, *Adv. Mater.* **2022**, *34*, 2205053.
- [66] M. Sajadi, M. Wolf, T. Kampfrath, *Opt. Express* **2015**, *23*, 28985.
- [67] G. Cataldo, J. A. Beall, H.-M. Cho, B. McAndrew, M. D. Niemack, E. J. Wollack, *Opt. Lett.* **2012**, *37*, 4200.
- [68] H. Kaveh, F. Jahangiri, T. Amini, *Opt. Continuum* **2022**, *1*, 388.
- [69] H. Xu, S. Kawata, D. J. Bergman, X. Zhu, W. Wei, X. Jiang, J. Nong, N. Chen, G. Lan, L. Tang, *Plasmonics II* **2016**, *125*, 10028.
- [70] F. Liu, E. Cubukcu, *Phys. Rev. B* **2013**, *88*, 115439.
- [71] Z. Zhang, J. Yang, T. Du, H. Ma, X. Jiang, *Carbon* **2022**, *190*, 80.
- [72] A. Bylinkin, M. Schnell, M. Autore, F. Calavalle, P. Li, J. Taboada-Gutiérrez, S. Liu, J. H. Edgar, F. Casanova, L. E. Hueso, P. Alonso-Gonzalez, A. Y. Nikitin, R. Hillenbrand, *Nat. Photonics* **2020**, *15*, 197.
- [73] K. Chen, R. Adato, H. Altug, *ACS Nano* **2012**, *6*, 7998.
- [74] C. Wu, A. B. Khanikaev, R. Adato, N. Arju, A. A. Yanik, H. Altug, G. Shvets, *Nat. Mater.* **2011**, *11*, 69.